MATLAB R2016a

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- #jungkook #jk.jpeg
- 0ebb83a1-c98b-4700-b...
- 1fef2246-0411-4ebf-a8...
- 8f9573c9-65b1-49b3-b...
- 24c81b04-5d55-4d62-8...
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- A01.m
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Details

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SASexp13.m

```
1 -   clc; clear; close all;
2 -   fm = 50;              % message (max) signal frequency, Hz
3 -   T  = 0.06;            % observation window, s
4 -   dt = 1e-5;            % fine grid to emulate CT signal
5 -   t  = 0:dt:T;
6 -   x  = cos(2*pi*fm*t);   % reference continuous-time signal
7
8 -   fs_list = [0.5*fm, fm, 2*fm, 5*fm, 10*fm];
9 -   labels  = {'fs = 0.5 fm (aliased)','fs = fm (under-sampled)', ...
10        'fs = 2 fm (Nyquist rate)','fs = 5 fm (over-sampled)','fs = 10 fm (over-sampled)'};
11 -  figure('Name','Sampling at Different Rates');
12 -  for k = 1:length(fs_list)
13 -      fs = fs_list(k);
14 -      n  = 0:1/fs:T;
15 -      xn = cos(2*pi*fm*n);
16 -      subplot(length(fs_list),1,k);
17 -      plot(t,x,'b-','LineWidth',1); hold on;
18 -      stem(n,xn,'r','filled');
19 -      title(labels{k}); xlabel('t (s)'); ylabel('x(t)');
    end
```

Command Window

```
Nyquist rate (2*fm) = 100.00 Hz
fs =   25.0 Hz (fs/fm =  0.5) -> mean reconstruction error = 1.06453
fs =   50.0 Hz (fs/fm =  1.0) -> mean reconstruction error = 1.04923
fs =  100.0 Hz (fs/fm =  2.0) -> mean reconstruction error = 0.20096
fs =  250.0 Hz (fs/fm =  5.0) -> mean reconstruction error = 0.03780
fs =  500.0 Hz (fs/fm = 10.0) -> mean reconstruction error = 0.01628
fx >>
```

Workspace

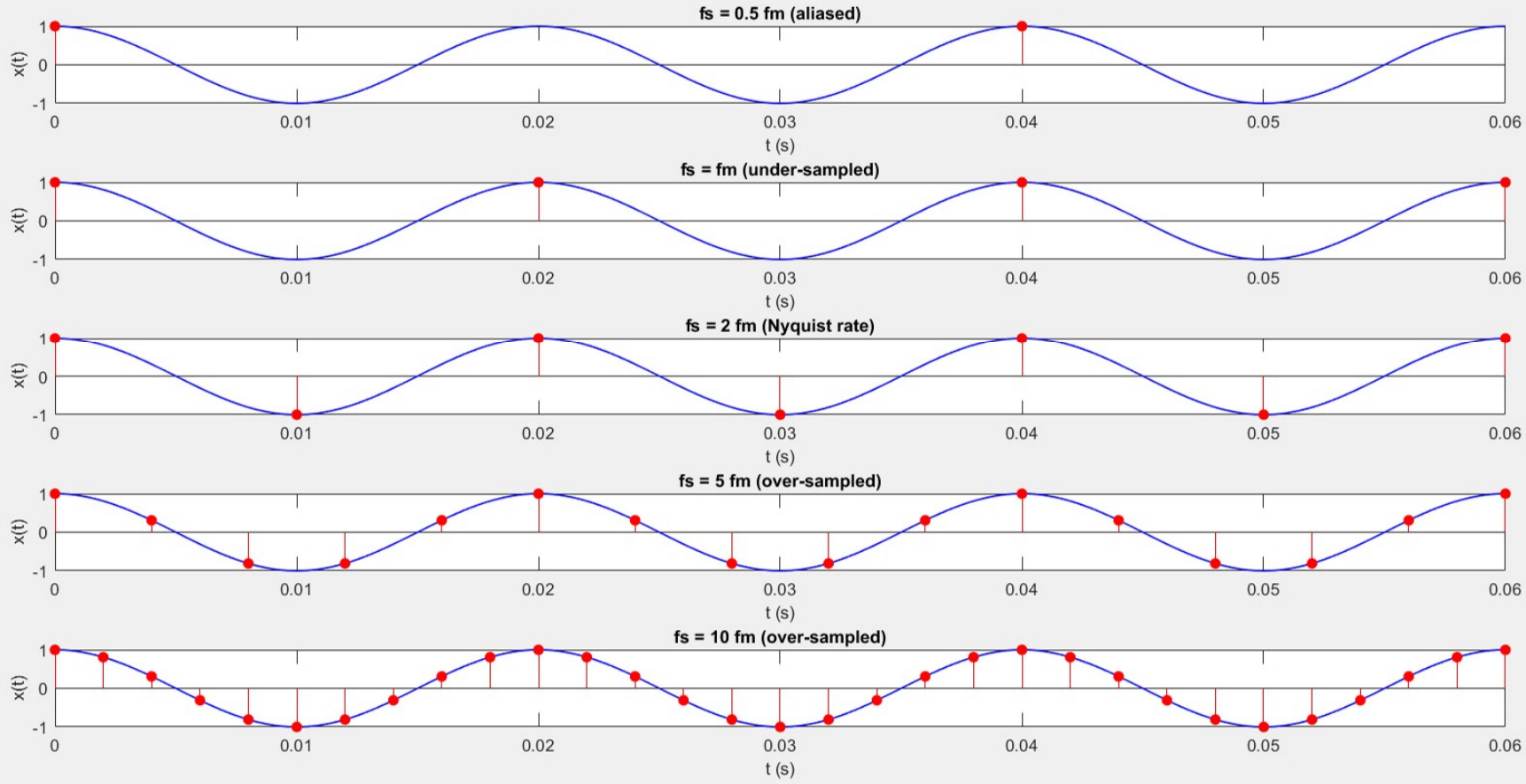
Name	Value
dt	1.0000e-05
err	[1.0645,1.0492,0....
fm	50
fs	500
fs_list	[25,50,100,250,50...
i	31
k	5
labels	1x5 cell
n	1x31 double
t	1x6001 double
T	0.0600
x	1x6001 double
x_recon	1x6001 double
xn	1x31 double

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Figure 1: Sampling at Different Rates

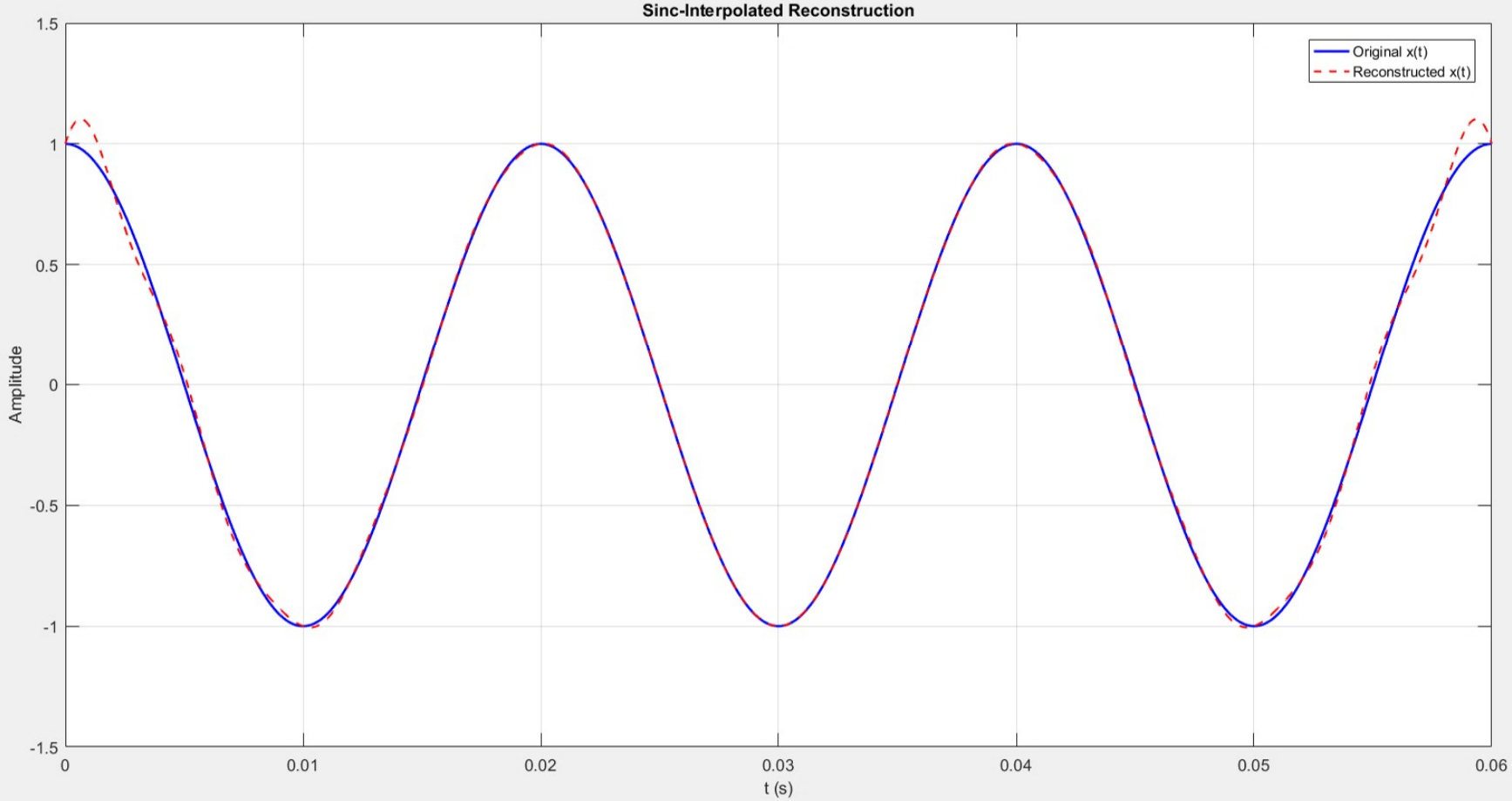
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Figure 2: Reconstruction at fs = 10 fm

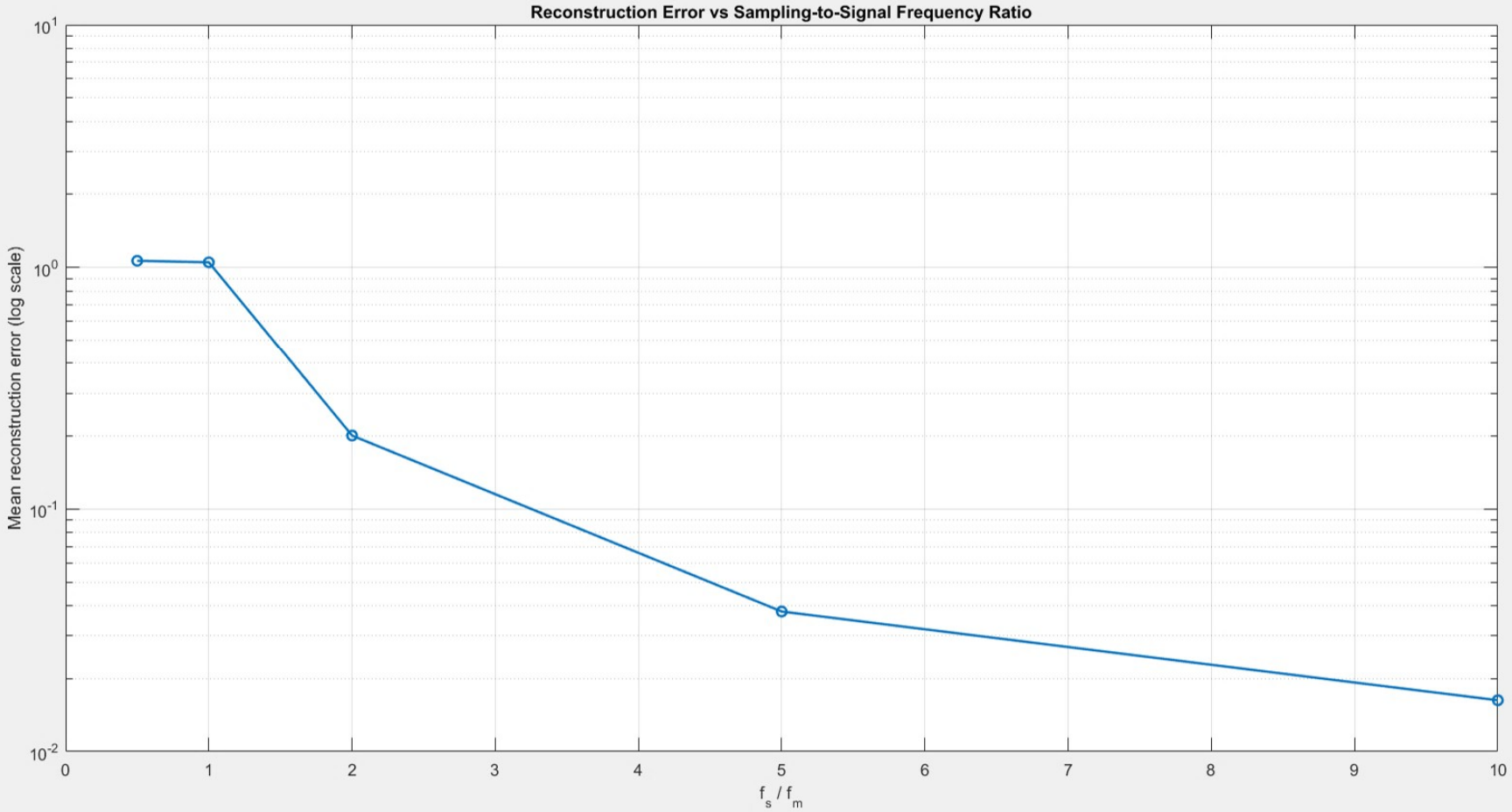
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Figure 3: Reconstruction Error vs Sampling Ratio

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